

Causal Evaluation Hierarchy

Strict dependency: success at Level N requires Level $N-1$

L1 State Identification

Interpreting current machine states and detecting anomalies from sensor data.

L2 Intervention Reasoning

Predicting the immediate impact of control changes or faults at the present timestep.

L3 Counterfactual Analysis

Simulating alternative historical timelines to assess hypothetical past conditions.

L4 Decision Support

Synthesizing multi-step recovery procedures using sensor fusion and technical manuals.

Physical Causal Schema



Healthy: $E = f(S, C)$ | Fault: $E \neq f(S, C)$

Fault $\Leftrightarrow$ empirical deviation in Effort given Setpoint and Context

Scalable Q&A Generation Engine

+80 Expert Templates

Expert-authored
across all 4 levels

Industrial Datasets

FactoryWave · AURSAD · Simulated Robotic
Real + simulated sensor streams

Dynamic Variable Sampling

Deterministic ground truth · Vendi Score validates diversity

Multi-Select

Ranking

Tensor Prediction

Open answers

Comparison with Existing Benchmarks

Benchmark	Robot	Multi. TS	Size	Counter.	Rank	Num.	Free	Novel TS
TimeSeriesExam	×	×	~700	×	✓	×	×	×
ChatTS	×	✓	~134k	×	✓	✓	×	×
EngineMT-QA	✓	✓	~110k	×	✓	✓	✓	×
TSAQA	Part.	×	~210k	×	✓	×	×	×
Time-MQA	×	✓	~200k	×	✓	✓	✓	×
MTBench	×	✓	?	×	✓	✓	✓	×
QuAnTS	×	✓	~150k	×	✓	✓	✓	×
CLEVR	×	×	~1M	×	×	✓	×	×
CLEVRER	×	×	~305k	✓	×	✓	×	×
FactoryBench	✓	✓	TBD	✓	✓	✓	✓	✓

The only benchmark guaranteeing verifiable truth
across deep causal temporal reasoning on machine data.